



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
13-06-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)

PAPER-2

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

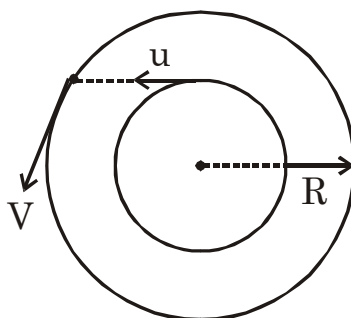
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 21)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
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Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Two particles are moving in concentric circular paths with constant speed u and V . Radius of outer circle is R . Distance between two particles is constant. Find the constant distance between two particles.



(A) $R\sqrt{1 - \frac{u}{V}}$

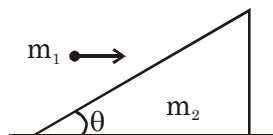
(B) $R\sqrt{1 - \frac{u^2}{V^2}}$

(C) $\sqrt{R^2 \left(1 - \frac{V}{u}\right)}$

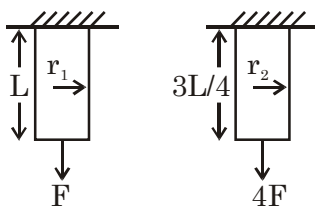
(D) $\sqrt{R^2 \left(1 - \frac{V^2}{u^2}\right)}$

Space for Rough Work

2. A wedge of mass m_2 is placed on a smooth horizontal surface. This wedge is struck by an elastic ball of mass m_1 whose velocity is horizontal just before impact. The ball bounces off the wedge and then lands on it again at the point of first impact. Value of $\frac{m_1}{m_2}$ is ($\theta < 45^\circ$)



- (A) $\cot^2 \theta - 1$ (B) $\cot^2 \theta + 1$ (C) $\tan^2 \theta - 1$ (D) $\tan^2 \theta + 1$
3. Two wires shown in diagram are made up of same material. If breaking force for first wire of radius r_1 is F and for second wire of radius r_2 is $4F$. Find $\frac{r_2}{r_1}$.



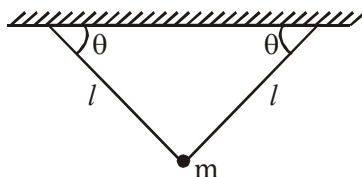
- (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 1 : 4

Space for Rough Work

4. Two neutral metal sphere of radius r each are connected by a light flexible conducting string of length L . The spheres are placed in a uniform field E directed along the line connecting the centre of spheres. Initially spheres are at a distance l from each other. ($r \ll l < L$). Choose **CORRECT** statement regarding the force of attraction between spheres.
- (A) Force is inversely proportional to distance between them.
 (B) Force is inversely proportional to square of distance between them.
 (C) Force is independent of distance between them.
 (D) Force is directly proportional to distance between them.
5. A non conducting wire of length l having uniform charge density λ C/m from $x = \frac{l}{2}$ to $x = l$ (Take x axis is along its length) is rotated with constant angular velocity ω about a line passing through its one end and perpendicular to its length. Magnetic field at fixed end is (at $x = 0$)
- (A) $\frac{\mu_0 \lambda \omega}{4\pi}$ (B) $\frac{\mu_0 \lambda \omega}{4\pi} \log_e 2$ (C) $\frac{\mu_0 \lambda \omega}{4}$ (D) $\frac{\mu_0 \lambda \omega}{4\pi^2} \log_e 2$

Space for Rough Work

6. A particle of mass m is in equilibrium in a vertical plane as given in diagram. If one string snaps then component of initial acceleration of particles in vertical direction is



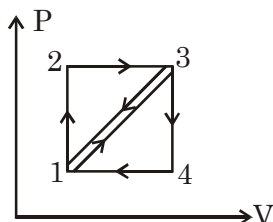
- (A) $g \sin \theta \cos \theta$ (B) $g \sin^2 \theta$ (C) $g \cos^2 \theta$ (D) $g \tan \theta$
7. We have two alike thin plano-convex glass lenses, with refractive index 1.5. The plane face of one of them and curved surface of the other is silver coated. What is ratio of their focal lengths.
- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 28)

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Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

8. Two samples of ideal gas, of the same amount are taken through cyclic process 1-2-3-1 and 1-3-4-1. Let efficiency of cyclic process 1-2-3-1 is η_1 and that of 1-3-4-1 is η_2 . Choose **CORRECT** statement.



(A) $\eta_1 < \eta_2$

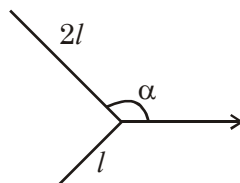
(B) $\frac{1}{\eta_1} - \frac{1}{\eta_2} = 1$

(C) $\eta_2 = \frac{\eta_1}{1 + \eta_1}$

(D) All of above are incorrect

Space for Rough Work

9. A straight metal rod of length $3l$ is rigidly bent through right angle as shown. The bent rod is then placed on a rough horizontal surface. A light string is attached to the vertex of the right angle. The string is then pulled horizontally so that rod slides at a constant velocity. Choose **CORRECT** statement(s).



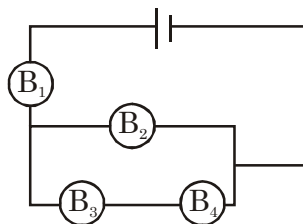
- (A) If frictional force on short leg is f then on longer leg frictional force is $2f$.
- (B) Angle α shown in diagram is $\pi - \tan^{-1} \frac{1}{4}$.
- (C) Angle α shown in diagram is $\tan^{-1} \frac{1}{4}$.
- (D) Angle α shown in diagram is $\frac{\pi}{2} + \tan^{-1} \frac{1}{4}$.
10. A source of sound S is moving with constant speed u_s and an observer is moving with constant speed u_o . Speed of sound in air is C ($C > u_o > u_s$) original frequency of source is f . Air is stationary. Choose **CORRECT** statement(s):-



- (A) Speed of sound recorded by observer is less than C .
- (B) Recorded frequency will be less than actual frequency.
- (C) Recorded frequency does not depend upon separation between source and observer.
- (D) Recorded wavelength will be different from actual wavelength.

Space for Rough Work

11. Four identical bulbs are connected as shown in the figure to a battery. Choose **CORRECT** statement(s) :

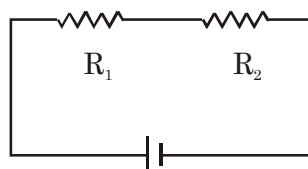


- (A) Bulb 1 illuminates most strongly.
 (B) Bulb 3 and 4 illuminate with equal intensity.
 (C) Bulb 2 illuminates more brightly as compared to bulb 3.
 (D) All bulb illuminates with equal brightness.
12. A certain planet of radius R is composed of a uniform material that through radioactive decay, generates a net power P . This results in a temperature difference between the inside and outside of the planet as heat is transferred from interior to the surface. The rate of heat transfer is governed by thermal conductivity which is constant for this planet and is equal to k . Assume that planet is in steady state

- (A) Surface temperature of planet is $T = \left(\frac{P}{4\pi\sigma R^2} \right)^{\frac{1}{4}}$ where σ is Stefan's constant.
 (B) Temperature difference between centre and surface of planet is $\Delta T = \frac{P}{8\pi kR}$.
 (C) Temperature difference between centre and surface of planet is $\Delta T = \frac{P}{2\pi kR}$
 (D) Temperature difference between centre and surface of planet is $\Delta T = \frac{P}{\pi kR}$

Space for Rough Work

13. A conducting sphere of radius R has a charge Q . A particle carrying a charge q is placed at a distance $2R$ from the surface of sphere. Choose **CORRECT** statement(s) :-
- (A) Even when a spherical metal surface has non uniform charge, the potential at centre is still due to sum of all elemental charges on conducting surface.
- (B) Potential at a point at a distance $R/2$ from centre is $\frac{q}{12\pi\epsilon_0 R} + \frac{Q}{4\pi\epsilon_0 R}$.
- (C) Potential at centre of sphere is $\frac{q}{12\pi\epsilon_0 R} + \frac{Q}{4\pi\epsilon_0 R}$.
- (D) Charge distribution on the surface of conducting sphere is non uniform.
14. The circuit shown in diagram contains an ideal battery of emf 6.0 V and two resistors R_1 and R_2 . A voltmeter is used across R_1 , its reading is 2.0 V . When same voltmeter is used across R_2 its reading is 3.0 V . Choose **CORRECT** option(s) :



- (A) Actual voltage across R_1 is 2.4 V . (B) Actual voltage across R_1 is 2.8 V .
 (C) Actual voltage across R_2 is 3.6 V . (D) Actual voltage across R_2 is 3.2 V .

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
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Paragraph for Questions 15 and 16

Although photons lack rest mass, photons behave as though they have gravitational mass.

Photon mass $m = \frac{p}{c} = \frac{h\nu}{c^2}$. All photons travel at the speed of light. However when photon falls through a height H near the surface of earth, then there is increase in its frequency. Because the frequency change is extremely small, we can neglect corresponding change in photon's mass $h\nu/c^2$.

Final photon's energy = initial photons energy + increase in its energy

$$\Rightarrow h\nu' = h\nu + mgH$$

$$= h\nu + \frac{h\nu}{c^2} gH$$

$$\Rightarrow h\nu' = h\nu \left(1 + \frac{gH}{c^2} \right)$$

Now answer the following questions.

15. Find change in frequency of a photon of red light whose original frequency is 7.3×10^{14} Hz when it falls through 22.5 m near surface of earth.
 (A) 1.8 Hz (B) 180 Hz (C) 1.8×10^3 Hz (D) zero
16. A photon is at surface of earth. Its frequency is ν . Mass of earth is M and radius is R . When photon reaches at infinity its frequency is

- (A) $\nu \left(1 - \frac{GM}{c^2 R} \right)$ (B) $\nu \left(1 + \frac{GM}{c^2 R} \right)$ (C) $\frac{\nu}{1 + \frac{GM}{c^2 R}}$ (D) $\frac{\nu}{1 - \frac{GM}{c^2 R}}$

Space for Rough Work

Paragraph for Questions 17 and 18

A screw gauge with pitch 1 mm and 100 divisions on circular scale is used to measure the thickness of a glass slab. When there is no object between the studs and its face touch each other, the 40th division of circular scale coincide with the reference line and zero of scale is not visible. When glass slab is held between the studs and reading of linear scale is found to be 5 and that of circular scale is 26.

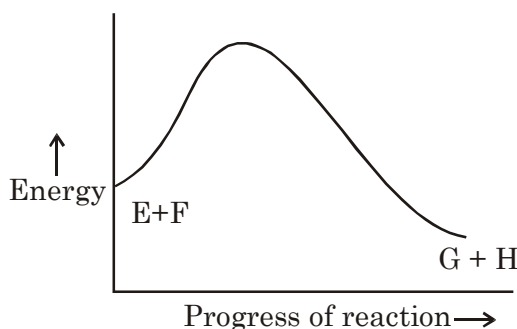
17. Zero error in the above mentioned screw gauge is
(A) -0.40 mm (B) -0.80 mm (C) -0.60 mm (D) -0.20 mm
18. Thickness of slab is
(A) 5.26 mm (B) 5.66 mm (C) 5.86 mm (D) 5.16 mm

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks : 21)**

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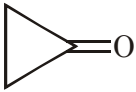
1. Consider the reaction $E + F \rightarrow G + H$ which has the following reaction coordinate diagram :



Find the correct statement ?

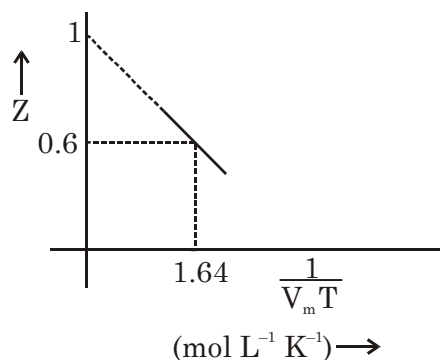
- (A) The activation energy is greater for forward reaction than backward reaction.
- (B) Percentage increase in rate is more for forward reaction than backward reaction, when temperature increases.
- (C) Reaction is endothermic.
- (D) Fraction of activation molecule is higher for forward reaction than backward reaction.

Space for Rough Work

2. Which of the following statement is **INCORRECT** ?
- (A) When aldehydes react with NaHSO_3 , sulphur acts as nucleophile.
- (B)  forms stable hydrate.
- (C) $\text{Ag}(\text{NH}_3)_2^+$ can be used to convert $\text{CH}_3 - \text{CH} = \text{CH} - \overset{\text{O}}{\parallel}{\text{C}} - \text{H}$ into $\text{CH}_3 - \text{CH} = \text{CH} - \overset{\text{O}}{\parallel}{\text{C}} - \text{O}^-$.
- (D) α -hydroxy ketones give negative tollens test.
3. Which of the following element/compound exhibits ion-dipole interaction with water molecules in aqueous solution?
- (A) Xe (B) HCl (C) NF_3 (D) Glucose
4. A solution contains 0.01 M concentration each of Cl^- , Br^- and I^- . $\text{AgNO}_3(\text{s})$ is slowly added to container. The % of Br^- remaining in the solution, when precipitation of AgCl starts is.
- (Given : $K_{\text{sp}}(\text{AgCl}) = 10^{-10}$, $K_{\text{sp}}(\text{AgBr}) = 10^{-12}$, $K_{\text{sp}}(\text{AgI}) = 10^{-18}$)
- (A) 0.5 (B) 2 (C) 1.5 (D) 1

Space for Rough Work

5. Which of the following is not used in synthesis of Aniline from nitrobenzene ?
 (A) H_2 (excess)/Pt (B) Sn/HCl (C) Fe/HCl (D) LiAlH_4
6. Which of the following compound exhibit partial hydrolysis when treated with excess of water?
 (A) PCl_3 (B) XeF_6 (C) SiF_4 (D) CaC_2
7. For a Vanderwaal gas 'X', the volume occupied by molecules is negligible. Its Z vs $\frac{1}{T V_m}$ graph was obtained as below. If the value of its vander Waal's constant, $a = y \times 10^{-2} \text{ L}^2 \text{ atm/mol}^2$, what is the value of y ? ($R = 0.082 \text{ L-atm/K-mol}$)



- (A) 1 (B) 3 (C) 4 (D) 2

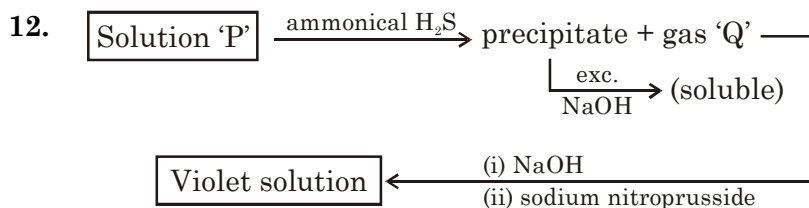
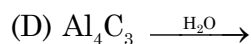
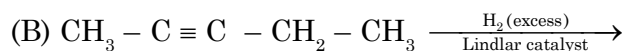
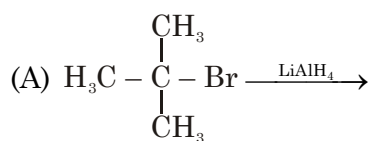
Space for Rough Work

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8. In amylopectin, which of the following type(s) of glycosidic linkage(s) is/are present ?
 (A) $\alpha - (1,2)$ (B) $\alpha - (1,4)$ (C) $\alpha - (1,6)$ (D) $\beta - (1,4)$
9. Which of the following anions produce gas/volatile compound(s) on treatment of their aqueous solution with conc. H_2SO_4 ?
 (A) CO_3^{2-} (B) Br^- (C) PO_4^{3-} (D) $\text{C}_2\text{O}_4^{2-}$
10. The voltaic cell using Cu/Cu^{2+} and Sn/Sn^{2+} half cells is set up at standard conditions and each compartment has a volume of 250 mL. The cell delivers 0.965 ampere for 10 hr, then
 (Given - atomic mass of Cu = 64 g/mol. $E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34\text{V}$ $E_{\text{Sn}^{2+}/\text{Sn}}^0 = -0.144\text{V}$)
 (A) Final molarity of Cu^{2+} in solution is 0.28 M
 (B) Final molarity of Sn^{2+} in solution is 0.72 M
 (C) Initial weight of Cu^{2+} in solution is 16 g.
 (D) Final amount of Cu^{2+} in solution is 4.48 g.

Space for Rough Work

11. In which of the following reaction(s), alkane is produced ?



Solution 'P' may contain which of the following salt?

- (A) CrCl_3 (B) ZnSO_4 (C) AlCl_3 (D) CaCl_2

13. Which of the following speices produce their oxy acid in highest oxidation state on treatment with con. HNO_3 ?

- (A) P_4 (B) I_2 (C) S_8 (D) HI

14. K_2MnO_4 is converted into KMnO_4 by treatment with which of the following compounds?

- (A) O_3 (B) $\text{CO}_2(\text{aq.})$
 (C) Cl_2 (D) Highly concentrated aq. solution of KOH

Space for Rough Work

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Paragraph for Questions 15 and 16

Acetic acid forms dimer in benzene. A solution containing 120 gram acetic acid in 2 kg of benzene has its freezing point depressed by 3.84 K. (Given : $K_f(\text{C}_6\text{H}_6) = 5.12 \text{ K kg mol}^{-1}$)

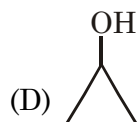
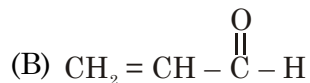
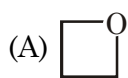
15. Initial molality of benzene solution in terms of acetic acid will be :
(A) 0.5 (B) 2 (C) 1 (D) 2.5
16. The degree of dimerization of acetic acid will be :
(A) 0.75 (B) 0.5 (C) 0.25 (D) 1

Space for Rough Work

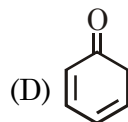
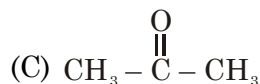
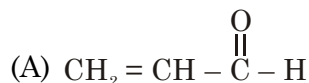
Paragraph for Questions 17 and 18

Tautomerism is a type of structural isomerism. Due to tautomerism a structural isomer can convert into another one. Tautomerism is either acid or base catalyzed. Although there are many types of tautomerism but keto-enol tautomerism is most common. Generally keto form is most stable but due to gain in aromaticity or hydrogen bonding, enol form may also be dominating. Temperature, solvent & steric hindrance can also affect percentage enol content.

17. Which of the following is tautomer of $\text{CH}_3 - \text{CH} = \text{CH} - \text{OH}$?



18. Which of the following compound does not undergo nucleophilic addition reaction when treated with grignard reagent ?



Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 21)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The equation $\sin x = [1 + \sin x] + [1 - \cos x]$ has 'k' number of solutions in $[0, 10\pi]$ then k is equal to (Here $[\cdot]$ represents greatest integer function)
 (A) 0 (B) 10 (C) 11 (D) 20
2. Value of $\cos^{-1}(\sin(-5)) + \tan^{-1}(\cot 7) - \sec\left(\cos^{-1}\left(\frac{-1}{3}\right)\right) + \sin(\operatorname{cosec}^{-1}2)$ is
 (A) $\frac{2\pi-15}{2}$ (B) $\frac{31-6\pi}{2}$ (C) $\frac{2\pi+3}{2}$ (D) $\frac{10\pi-17}{2}$
3. A triangle is formed by the lines $x + y = 0$, $x - y = 0$ and $lx + my = 1$. If l and m vary subject to the condition $l^2 + m^2 = 1$, then the locus of its circumcentre is
 (A) $2(x^2 + y^2) = x + y$ (B) $2(x^2 + y^2) = x + 2y$
 (C) $4(x^2 + y^2) = 2x + y$ (D) $2(x^2 - y^2) = x + y$

Space for Rough Work

4. Let the circle $x^2 + y^2 + 2gx + 2fy + 3 = 0$ meets the rectangular hyperbola $xy = c^2$ at four points A, B, C, D. The equation of circumcircle of the triangle formed by joining orthocentre of the triangles ABC, ABD and ACD is $x^2 + y^2 + mx + ny + p = 0$. If $g + f + 9 = 0$, then value of $m + n + p$ is
- (A) 9 (B) 15 (C) 21 (D) 24
5. Let α, β, γ be roots of the equation $z^3 - 2z^2 + 3z - 4 = 0$, then value of $\frac{1}{\alpha\beta + \gamma - 1} + \frac{1}{\beta\gamma + \alpha - 1} + \frac{1}{\gamma\alpha + \beta - 1}$ is
- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$
6. Among 20 people, 8 people are infected with a virus named as COVID. Number of ways in which we can form 4 equal groups of 20 people so that people infected with COVID should restrict to 2 groups only is $k \times {}^9C_5 \times {}^{10}C_5$, take 'k' is
- (A) 8 (B) 20 (C) 33 (D) None of these
7. Sum of rational terms in the expansion of $(\sqrt{2} + \sqrt[3]{3} + \sqrt[6]{7})^{10}$ is
- (A) 7560 (B) 5880 (C) 13440 (D) 13472

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 28)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

8. $f(x) = \min \phi(t), -3 \leq t \leq x$, where $\phi(t) = ||x-1| - |x+1||$ then
- (A) $f(x)$ is non-differentiable at $x = 0, -1$ (B) $f(x)$ is non-differentiable at $x = -1, 1$
- (C) $f'(100) = 0$ (D) $\int_{-3}^{10} f(x) dx = 5$
9. Let $f(x) = \begin{cases} ax^2 + bx + c, & x < 1 \\ 2ax + b, & x \geq 1 \end{cases}$ where $a \neq 0$. If $f(x)$ is differentiable at $x = 1$, then
- (A) Minimum of f is a , when $a > 0$
- (B) Maximum of f is a , when $a < 0$
- (C) f increases, when x and a have same sign
- (D) f increases, when $0 < x < 1$

Space for Rough Work

10. If $k = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x}{x} dx$, then

(A) $k \in \left(\frac{1}{2}, 1\right)$

(B) $k \in \left(\frac{\sqrt{3}}{4}, \frac{1}{2}\right)$

(C) $k \in \left(0, \frac{1}{2}\right)$

(D) $k \in \left(\frac{\sqrt{3}}{4}, \frac{\pi}{6}\right)$

11. The function $f(x)$ satisfies the functional equation $f(\tan \theta) = \frac{\sin^2 2\theta}{4}$ for all real θ . The area

enclosed by $f(x)$ with x axis is given by $\frac{a\pi}{b}$, where 'a' and 'b' are coprime natural numbers.

Now mark the correct statement(s).

(A) $2b + a$ is a prime number

(B) b is a prime number

(C) $2a + b$ is a prime number

(D) $a + b$ is a prime number

Space for Rough Work

12. Which of the following is true for the equation $Z^2 = Z + \frac{2}{|Z|^3}$ where $Z = x + iy$?

(A) Number of complex numbers satisfying the equation is 3.

(B) Number of complex numbers satisfying the equation is 4.

(C) Sum of modulus of all complex numbers satisfying the equation is $2 + 2^{\frac{5}{4}}$.

(D) Product of modulus of all complex numbers satisfying the equation is $\sqrt{2}$.

13. Let $T_r = \frac{4r^2 - 2}{4r^4 + 1}$ and $\sum_{r=1}^n T_r = 1 - S_n$, then mark the correct option(s).

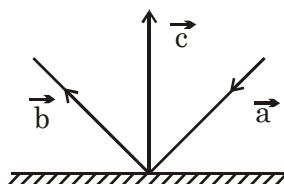
(A) $\lim_{n \rightarrow \infty} nS_n = 2$

(B) $\lim_{n \rightarrow \infty} nS_n = 1$

(C) $S_{10} = \frac{21}{221}$

(D) $S_5 = \frac{11}{61}$

14. In the adjacent figure $\vec{a}$ is along incident ray and $\vec{b}$ reflected ray, $\vec{c}$ is the vector perpendicular to the mirror, then which of the following is correct if all vectors are unit vector ?



(A) $\vec{c} = \frac{\vec{b} - \vec{a}}{|\vec{b} - \vec{a}|}$

(B) $\vec{b} = \vec{a} + \vec{c}|\vec{b} - \vec{a}|$

(C) $\vec{b} = \vec{a} - \vec{c}$

(D) $\vec{c} = \vec{a} + \vec{b}$

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

If a circle with centre C (α, β) intersects a rectangular hyperbola with centre L (h, k) at four points P(x_1, y_1), Q(x_2, y_2), R(x_3, y_3), S(x_4, y_4), then the mean of the four points P, Q, R, S is the mean, of the points C and L. In other words the midpoint of CL coincides with the mean point of P, Q, R, S.

$$\text{Analytically } \frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{\alpha + h}{2}, \frac{y_1 + y_2 + y_3 + y_4}{4} = \frac{\beta + k}{2}$$

Choose the correct answer

15. Five points are selected on a circle of radius a . The centres of rectangular hyperbolas, each passing through four of these points lie on a circle of radius.
- (A) a (B) $2a$ (C) $\frac{a}{\sqrt{2}}$ (D) $\frac{a}{2}$
16. A circle with fixed centre ($3h, 3k$) and of variable radius cuts the rectangular hyperbola $x^2 - y^2 = 9a^2$ at the points A, B, C, D. The locus of the centroid of the ΔABC is given by
- (A) $(x - 2h)^2 - (y - 2k)^2 = a^2$ (B) $(x - h)^2 - (y - k)^2 = a^2$
- (C) $\frac{x^2}{h^2} - \frac{y^2}{k^2} = a^2$ (D) $\frac{x^2}{h^2} + \frac{y^2}{k^2} = a^2$

Space for Rough Work

Paragraph for Questions 17 and 18

For different values of γ , the plane $(1 + \gamma)x + (2 - \gamma)y + (1 - \gamma)z - (1 + 2\gamma) = 0$ passes through a common line L. Based on this information, answer the following questions.

17. The value of ' γ ' so that plane contains line L and parallel to line $\frac{x-1}{3} = \frac{1-y}{1} = \frac{z+1}{2}$ is
- (A) 2 (B) -1 (C) $-\frac{3}{2}$ (D) None of these
18. If plane $x + 2y + z = 1$ is rotated about line L by an angle of 90° , then value of γ is
- (A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	7	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	21
I(ii)	One or more correct option(s)	7	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–1 In all other cases	28
I(iii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 In all other cases	—	12

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator